

1 2025-01-13 | Week 1 | Lecture 1

This lecture is based on section 8.1 in the textbook.

1.1 Arc Length

First, recall the definition of derivative:

$$f'(x) := \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

or equivalently,

$$f'(x) = \lim_{\tilde{x} \rightarrow x} \frac{f(\tilde{x}) - f(x)}{\tilde{x} - x} =$$

In other words, the derivative is obtained by computing

$$\frac{\text{change in } y\text{-value}}{\text{change in } x\text{-value}} = \frac{\Delta y}{\Delta x}$$

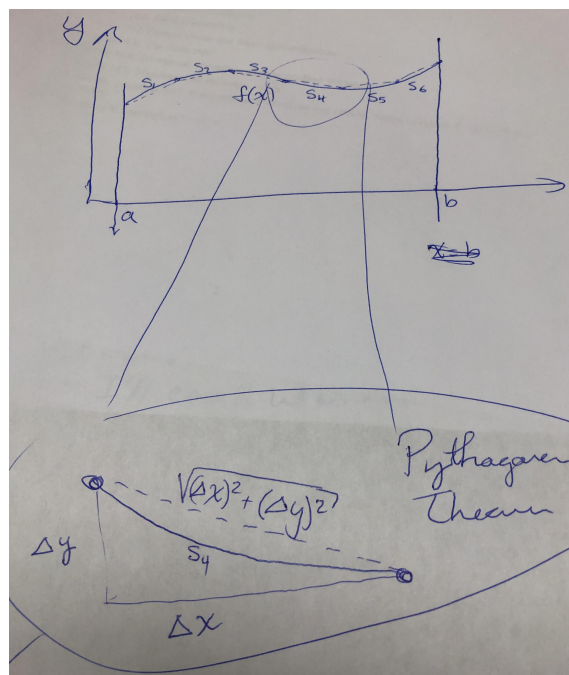
as $\Delta x \rightarrow 0$.

One of the central ideas in calculus is to approximate curvy things with flat things. For example, if faced with a curve, approximate it with straight lines. For example,

Problem: I am an ant, and I want to travel from a to b by walking on the graph of $f(x)$. How far must I walk?

Strategy: Cheat a little: think of the ant walking in a bunch of straight lines that are pretty close to the curve.

Here's the picture



Here,

$$L = s_1 + s_2 + \dots + s_6$$

where, for example,

$$\begin{aligned}
s_4 &\approx \sqrt{(\Delta x)^2 + (\Delta y)^2} && \text{since the line segment length approximates the curve length} \\
&= \sqrt{\left[1 + \left(\frac{\Delta y}{\Delta x}\right)^2\right] (\Delta x)^2} \\
&= \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2} \Delta x \\
&\approx \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx && \text{this holds when } \Delta x \approx 0, \text{ so long as the curve is smooth} \\
&= \sqrt{1 + [f'(x)]^2} dx && \text{by change of notation}
\end{aligned}$$

Theorem 1 (Arc length formula). *Let f be a differentiable function on the interval $[a, b]$, and assume that f' is continuous on $[a, b]$. Then the arc length of f from $x = a$ to $x = b$ is*

$$\text{Arc Length} = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

Example 2 (Arc length). **Problem:** Find the length of the curve defined by $f(x) = 2x^{3/2}$, where x ranges from $x = 0$ to $x = 2$.

Solution: $f'(x) = 3x^{\frac{1}{2}}$, so

$$\begin{aligned}
\text{Arc length} &= \int_0^2 \sqrt{1 + [f'(x)]^2} dx \\
&= \int_0^2 \sqrt{1 + 9x} dx
\end{aligned}$$

which we can solve with a u -substitution.

End of Example 2. $\square$

Example 3 (Arc length). **Problem:** Find the length of the curve defined by $f(x) = \frac{1}{2}(e^x + e^{-x})$ from $x = -1$ to $x = 3$.

Solution: We first claim that the following equality holds:

$$1 + [f'(x)]^2 = \left(\frac{e^x + e^{-x}}{2}\right)^2. \quad (1)$$

This equality isn't obvious... it takes some work to deduce, which is shown below:

Proof of Eq. (1). Observe that $f'(x) = \frac{1}{2}(e^x - e^{-x})$. Therefore

$$\begin{aligned}
 1 + [f'(x)]^2 &= 1 + \left[\frac{1}{2} (e^x - e^{-x}) \right]^2 \\
 &= 1 + \frac{1}{4} (e^{2x} - 2 + e^{-2x}) \\
 &= \frac{1}{4} e^{2x} + \frac{1}{2} + \frac{1}{4} e^{-2x} && \text{expanding everything out} \\
 &= \frac{1}{4} (e^{2x} + 2 + e^{-2x}) && \text{factor out } \frac{1}{4} \\
 &= \frac{1}{4} (e^x + e^{-x})^2 \\
 &= \left(\frac{e^x + e^{-x}}{2} \right)^2.
 \end{aligned}$$

This shows that Eq. (1) holds. □

Now with Eq. (1) in hand, everything simplifies really nicely when we apply the formula from Theorem 1:

$$\begin{aligned}
 \text{Arc Length} &= \int_{-1}^3 \sqrt{1 + [f'(x)]^2} dx \\
 &= \int_{-1}^3 \sqrt{\left(\frac{e^x + e^{-x}}{2} \right)^2} dx \\
 &= \int_{-1}^3 \frac{e^x + e^{-x}}{2} dx \\
 &= \frac{1}{2} \int_{-1}^3 (e^x + e^{-x}) dx \\
 &= \frac{1}{2} [e^x - e^{-x}]_{x=-1}^{x=3} \\
 &= \frac{1}{2} [(e^3 - e^{-3}) - (e^{-1} - e^1)] \\
 &\approx 11.2
 \end{aligned}$$

End of Example 3. □

Tips

- Check that your answer is positive. Arc length is never negative.
- These problems are usually contrived so that you can actually do the integral. If the integral looks took hard, double check your work. Also, there might be a trick to the integral.